

# Exact height-one probabilities of three-dimensional Abelian sandpiles, spanning-tree entropies of the snub-square and Cairo lattices, and elementary resistor-network reductions

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We compute exact closed forms for three related lattice quantities: the probability  $P_1$  that a site of the Abelian sandpile has minimal height, the spanning-tree entropy, and two-point resistances. All three are controlled by the same Kirchhoff transfer currents. In three dimensions we obtain  $P_1$  for the simple-cubic, body-centred cubic, face-centred cubic and pyrochlore lattices as polynomials in the corresponding Watson integral  $W$  and in  $1/(\pi^2 W)$ . For example, on the simple cubic lattice  $P_1 = \frac{4}{3}\alpha^3(2 - 3\alpha)^2 = 0.054582470221\dots$  with  $\alpha = (7W + 54/(\pi^2 W))/36$ . For bipartite lattices of girth at least 6 with symmetric second neighbours we prove  $P_1 = (z-2)^{z-1}/[z(z-1)^{z-1}]$ , which gives  $1/12$  for the honeycomb (known) and the Laves graph, and  $2/27$  for diamond. In two dimensions we give exact  $P_1$  for the king, 4.8.8, Union Jack, star, dice, Lieb, snub-square, checkerboard, triakis and next-nearest-neighbour honeycomb lattices. The spanning-tree entropy of the snub-square (Shastry–Sutherland) lattice is  $z = \frac{10}{3\pi}G + \frac{1}{3}\ln(2 + \sqrt{3})$ , and that of its dual, the Cairo pentagonal lattice, is  $\frac{20}{9\pi}G + \frac{2}{9}\ln(2 + \sqrt{3})$ , where  $G$  is Catalan’s constant. The Shastry–Sutherland network with arbitrary dimer conductance has elementary resistances and a Clausen-function entropy. The king lattice has  $z = 20G/(3\pi)$ . We give a single elementary formula for the axis-bond resistance of any square lattice with diagonal bonds, and we reduce next-nearest-neighbour networks on bipartite girth- $\geq 6$  lattices to one Green function (elliptic for the honeycomb, Joyce’s FCC form for diamond). Every result was identified by PSLQ at 25–60 digits and confirmed independently by exact finite-volume computations or Monte Carlo simulation. An appendix gives a Hurwitz-zeta closed form for the added mass of a sphere touching the inside of a spherical container.

## I. INTRODUCTION

The Abelian sandpile model [1, 2] is the standard exactly solvable model of self-organized criticality. Majumdar and Dhar showed that the probability  $P_1$  that a site has the minimal height in the stationary state is a finite determinant of lattice Green functions [3]. On the square lattice this gives  $P_1 = 2/\pi^2 - 4/\pi^3$ . Priezzhev computed the higher heights [4]. Exact values have since been obtained on the honeycomb [5], triangular and hexagonal [6], and, very recently, kagome [7] lattices. The Majumdar–Dhar method applies in any dimension. Nevertheless, to our knowledge no closed form for  $P_1$  on a three-dimensional lattice has appeared. The obstacle is the need for exact resistances beyond nearest neighbours.

We remove that obstacle by identifying every needed resistance in the natural basis  $\{1, W, 1/(\pi^2 W)\}$  of the relevant Watson integral [8, 9]. The same machinery, based on Kirchhoff’s transfer currents [10] and their spanning-tree interpretation [11, 12], also gives new spanning-tree entropies. Numerical values for many Archimedean lattices are known from Chang and Wang [13], Chang and Shrock [14], and Guttman and Rogers [15]; we add closed forms. It also gives exact resistances on networks whose resistances were previously known only numerically [16] or via mappings [17].

*Status of novelty.* All statements were checked by web search against the literature. The honeycomb and

kagome values of  $P_1$  are known [5, 7]; we reproduce them as validation. The king-lattice entropy  $20G/(3\pi)$  may be implicit in the Mahler-measure literature [15]. For the remaining results our searches found no prior closed form. Novelty should be read in that limited sense.

## Notation

All resistors are  $1\Omega$  unless a conductance is specified, and  $R(x)$  is the effective resistance between the origin and  $x$ . For an arc  $a = (o \rightarrow h_a)$  the transfer current is  $Y_{ab} = \frac{1}{2}[R(h_a) + R(h_b) - R(h_a - h_b)]$ .  $G$  denotes Catalan’s constant, and  $\text{Cl}_2(\theta) = \sum_{n \geq 1} \sin(n\theta)/n^2$  is the Clausen function. Complete elliptic integrals use the parameter convention  $K(m) = \int_0^{\pi/2} (1 - m \sin^2 \theta)^{-1/2} d\theta$ .  $\mathbb{E}_k$  denotes the normalized Brillouin-zone average.

## II. ABELIAN SANDPILES ON 3D LATTICES

### A. Method

In the Abelian sandpile on a graph, a site  $v$  topples when its height exceeds  $\deg(v)$  and sends one grain to each neighbour. Majumdar and Dhar [3] showed that

$$P_1 = \det(I - M), \quad M_{jk} = D(x_j) + D(x_k) - D(x_j - x_k), \quad (1)$$

where  $D = R/2$  and  $x_j$  runs over the  $z - 1$  neighbours of the origin whose bonds are removed by the defect (one

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bond is kept). The matrix  $M$  is exactly the transfer-current matrix  $Y$  restricted to  $z - 1$  arcs. Hence

$$P_1 = \det(I - Y'), \quad (2)$$

with  $Y'$  equal to  $Y$  minus one arc. An internal consistency check is that  $P_1$  must not depend on which bond is kept. The check is nontrivial when the bonds at a site are inequivalent, and it holds for every result below.

*Corollary (UST leaves).* Since  $P_1$  is independent of the kept bond, the probability that a degree- $z$  vertex is a leaf of the uniform spanning tree is exactly  $zP_1$ . On the square lattice this reproduces the known value  $(8/\pi^2)(1 - 2/\pi)$ .

### B. Watson integrals

Table I lists the three constants we need. For FCC we use  $W_f = \mathbb{E}[1/(3 - s)]$ , which is one third of Watson's normalization  $\mathbb{E}[1/(1 - s/3)]$ .

### C. Resistances and results

In each case below,  $q \equiv 1/(\pi^2 W)$  with  $W$  the lattice's own Watson integral.

*Simple cubic* ( $z = 6$ ). The resistances are  $R(1, 0, 0) = 1/3$ ,  $R(1, 1, 0) = \alpha$  and  $R(2, 0, 0) = 2 - 4\alpha$ , with

$$\alpha = \frac{7W + 54/(\pi^2 W)}{36} = 0.395079152341851372 \dots \quad (3)$$

The  $5 \times 5$  determinant factorizes:

$$P_1^{\text{SC}} = \frac{4}{3} \alpha^3 (2 - 3\alpha)^2 = 0.054582470221053656104 \dots \quad (4)$$

*Body-centred cubic* ( $z = 8$ , neighbours  $(\pm 1, \pm 1, \pm 1)$ ). The resistances are  $R(1, 1, 1) = 1/4$ ,  $R(2, 0, 0) = (W - 4q)/4$ ,  $R(2, 2, 0) = 4q$  and  $R(2, 2, 2) = (8 - 3W - 36q)/4$ . Then

$$\begin{aligned} P_1^{\text{BCC}} &= \frac{3q}{128} (W + 4q)^3 (4 - W - 12q)^3 \\ &= 0.042454248763323867910 \dots \end{aligned} \quad (5)$$

*Face-centred cubic* ( $z = 12$ , neighbours are the permutations of  $(\pm 1, \pm 1, 0)$ ). The resistances are  $R(1, 1, 0) = 1/6$ ,  $R(2, 0, 0) = (4W - 3q)/6$ ,  $R(2, 1, 1) = (2W + 3q - 1)/3$  and  $R(2, 2, 0) = (8 - 12W - 9q)/3$ . Then

$$\begin{aligned} P_1^{\text{FCC}} &= 6^{-10} (5 - 4W - 6q)^2 (7 - 8W - 3q)^3 \\ &\quad \times (8W + 3q + 1)^3 (16W + 15q - 5)^3 \\ &= 0.029126520511690534744 \dots \end{aligned} \quad (6)$$

*Pyrochlore* ( $z = 6$ , corner-sharing tetrahedra,  $W = W_f$ ,  $X = 1/(\pi^2 W_f)$ ). The Laplacian transfer currents from an edge are  $1/3$  (self),  $1/6$  (the two other edges of the same tetrahedron),  $(1 + 8W + 3X)/48$  (collinear

partner) and  $(15 - 8W - 3X)/96$  (the other two). Then Eq. (2) gives a perfect square:

$$\begin{aligned} P_1^{\text{pyro}} &= \left[ \frac{(31 - 8W - 3X)(67 + 24W + 9X)}{9216} \right]^2 \\ &= 0.053582950138955615678 \dots \end{aligned} \quad (7)$$

*Diamond* follows from the general theorem of Sec. III A:  $P_1 = 2/27$ .

### D. Independent verification

We computed  $P_1$  *without sampling* at the centre of finite boxes with dissipative boundaries, by exact linear algebra (no sampling). The sequences converge to the formulas:

| Lattice  | Boxes            | Exact finite-box $P_1$          |
|----------|------------------|---------------------------------|
| SC       | $N = 16, 24, 32$ | 0.0546206, 0.0545941, 0.0545875 |
| BCC      | $N = 16, 24, 32$ | 0.0424902, 0.0424653, 0.0424590 |
| FCC      | $N = 12, 18, 24$ | 0.0291795, 0.0291427, 0.0291335 |
| FCC      | $N = 30$         | 0.0291301                       |
| Pyrochl. | $N = 10, 16, 22$ | 0.0536309, 0.0535949, 0.0535876 |
| Diamond  | $N$ increasing   | 0.0742235, 0.0741125, 0.0740859 |
| Laves    | $N = 10, 16, 24$ | 0.0834488, 0.0833615, 0.0833418 |

Monte Carlo simulations on  $32^3$ – $44^3$  lattices give SC 0.05451(23), BCC 0.04244(21) and FCC 0.02926(8). An earlier FCC run gave 0.02865(17), which we attribute to slow mixing; the exact finite-box sequence resolves the discrepancy in favour of the formula. The corresponding UST leaf probabilities  $zP_1$  are SC 0.3274948, BCC 0.3396340, FCC 0.3495182, pyrochlore 0.3214977 and diamond 8/27.

## III. EXACT $P(h = 1)$ CALCULATIONS

### A. A general theorem for bipartite lattices

*Theorem 1.* Let the lattice be  $z$ -regular, bipartite, of girth at least 6 (so that  $A^2 = zI + A_2$ ), and edge-transitive, and let its second-neighbour graph  $A_2$  also be edge-transitive. Then  $R(\text{nearest}) = 2/z$ ,  $R(\text{second}) = 2/(z - 1)$ , and

$$P_1 = \frac{(z - 2)^{z-1}}{z(z - 1)^{z-1}}. \quad (8)$$

*Proof.* Write  $L_2 = z^2 I - A^2 = LQ$ . This operator acts within each sublattice, so  $R_L(u, v) = z R_{L_2}(u, v)$  for  $u, v$  on the same sublattice. Foster's theorem [18] on the edge-transitive second-neighbour graph gives  $R_{L_2} = 2/[z(z - 1)]$  across a second-neighbour bond. Hence  $Y'$  has the constant diagonal  $2/z$  and constant off-diagonal entries. Its determinant then factorizes into Eq. (8).  $\square$

The theorem gives  $1/12$  for the honeycomb, which is known [5], and  $1/12$  for the Laves graph (srs). It

TABLE I. Watson integrals ( $c_i = \cos k_i$ ,  $s = c_1 c_2 + c_2 c_3 + c_3 c_1$ ).

| Lattice | Definition and closed form                                                                                                                                       | Value         |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| SC      | $\mathbb{E}\left[\frac{1}{1-(c_1+c_2+c_3)/3}\right] = \frac{\sqrt{6}\Gamma(\frac{1}{24})\Gamma(\frac{5}{24})\Gamma(\frac{7}{24})\Gamma(\frac{11}{24})}{32\pi^3}$ | 1.51638605915 |
| BCC     | $\mathbb{E}\left[\frac{1}{1-c_1 c_2 c_3}\right] = \frac{\Gamma(\frac{1}{4})^4}{4\pi^3}$                                                                          | 1.39320392969 |
| FCC     | $\mathbb{E}\left[\frac{1}{3-s}\right] = \frac{3\Gamma(\frac{1}{3})^6}{2^{14/3}\pi^4}$                                                                            | 0.44822039439 |

gives  $2/27 = 0.0740740\dots$  for diamond and applies to  $d$ -dimensional diamond with  $z = d + 1$ .

### B. The king lattice

Take  $\mathbb{Z}^2$  with all eight king-move neighbours ( $z = 8$ ). Put  $p = 1/\pi$  and  $L = \ln(2 + \sqrt{3})/(\pi\sqrt{3}) = 0.24202564754285515484$ . The resistances are

$$\begin{aligned} R(1,0) &= L, & R(1,1) &= \frac{1}{2} - L, \\ R(2,0) &= \frac{4}{\pi} - 4L, & R(2,1) &= 6L - \frac{1}{2} - \frac{2}{\pi}, \\ R(2,2) &= 5 - \frac{4}{\pi} - 14L, \end{aligned} \quad (9)$$

and the  $7 \times 7$  determinant factorizes as

$$\begin{aligned} P_1^{\text{king}} &= -\frac{1}{2048}(3 + 4p - 14L)(1 + 4p - 2L)(6L + 12p - 7) \\ &\quad \times (13 - 32p + 48p^2 - 52L - 12L^2)^2 \\ &= 0.0420560364703691189817\dots \end{aligned} \quad (10)$$

A Monte Carlo simulation on  $96 \times 96$  gives  $0.04218 \pm 0.00033$ .

### C. Further two-dimensional lattices

Table II collects the remaining exact values. We write  $c = \sqrt{2} \arccos(1/3)/\pi = 0.55412642397957198712$  for the  $4.8.8/\text{Union Jack}$  constant,  $p = 1/\pi$  and  $r = \sqrt{3}/\pi$ .

*Honeycomb with next-nearest neighbours* (unit weights,  $z = 9$ ). Let  $g = 4K(64/189)/(3\sqrt{21}\pi) = 0.16088977460548913890$ , the triangular Green function at  $t = 13/2$ , and let  $g' = dg/dt$ . All resistances among the site and its nine neighbours are rational combinations of  $1, r, g, g'$ , and

$$\begin{aligned} P_1 &= \frac{(2 - 7g)(12r - 14 - 105g - 1120g') \mathcal{P}^2}{196036848}, \\ \mathcal{P} &= 729120gg' + 1764gr^2 - 7224gr + 9163g + 8202600g'^2 \\ &\quad - 136080g'r + 305760g' + 792r^2 - 2688r + 2548, \end{aligned} \quad (11)$$

which evaluates to  $P_1 = 0.037899503465150422174$ . Exact finite boxes of  $40^2$ ,  $80^2$  and  $160^2$  cells give  $0.0379335$ ,  $0.0379082$  and  $0.0379017$ .

*Validation against the literature.* The same pipeline gives the kagome value  $(5 + 3r)(17 - 6r)/1296$ , which

matches the value reported in Ref. [7]. It gives  $1/12$  for the honeycomb [5] and reproduces the square lattice's  $2/\pi^2 - 4/\pi^3$  to 20 digits. Monte Carlo checks on open patches:  $4.8.8$   $0.0807\text{--}0.0817 (\pm 0.0006)$ ; star  $0.0697\text{--}0.0706 (\pm 0.0008)$ ; dice hub  $0.01273(30)$ , rim  $0.1888(10)$ ; Union Jack corner  $0.02024(30)$ , centre  $0.1535(8)$ . The checkerboard and triakis values were confirmed by exact finite boxes ( $40^2, 80^2, 160^2$  cells):  $0.0535660$ ,  $0.0535281$ ,  $0.0535185$  and  $0.0035988$ ,  $0.0035940$ ,  $0.0035928$ .

## IV. SPANNING-TREE ENTROPY ON CAIRO AND SNUB-SQUARE LATTICES

The number of spanning trees of an  $N$ -vertex piece of a lattice grows as  $e^{zN}$  [19]. For a periodic lattice,  $z$  is  $|V_{\text{cell}}|^{-1} \mathbb{E}_k \ln \det L(k)$ .

### A. Snub square and Cairo

For the snub square tiling  $3^2.4.3.4$  (the Shastri–Sutherland lattice, 4 vertices per cell),

$$\begin{aligned} \det L(k) &= 2(170 - 72(\cos k_1 + \cos k_2) - 28 \cos k_1 \cos k_2 \\ &\quad + \cos 2k_1 + \cos 2k_2). \end{aligned} \quad (12)$$

We find

$$\mathbb{E}_k \ln \det L = \frac{40}{3\pi}G + \frac{4}{3} \ln(2 + \sqrt{3}), \quad (13)$$

and hence

$$\begin{aligned} z_{\text{snub}} &= \frac{10}{3\pi}G + \frac{1}{3} \ln(2 + \sqrt{3}) \\ &= 1.41085564574433483667\dots, \end{aligned} \quad (14)$$

$$\begin{aligned} z_{\text{Cairo}} &= \frac{20}{9\pi}G + \frac{2}{9} \ln(2 + \sqrt{3}) \\ &= 0.94057043049622322445\dots \end{aligned} \quad (15)$$

The Cairo pentagonal tiling is the planar dual. It has the same number of trees per cell but 6 vertices per cell.

*Why it closes.* The  $z_2$ -discriminant of the spectral curve  $\det L = 0$  has only one simple factor,  $z_1^2 - 98z_1 + 1$ , whose roots are  $(\sqrt{3} \pm \sqrt{2})^4$ . After the node at  $k = 0$  is removed, the curve therefore has genus zero. The entropy must then be a combination of dilogarithms at algebraic points.

TABLE II. Exact height-one probabilities  $P_1$  of the Abelian sandpile. Each value is independent of which bond is kept in Eq. (1).

| Lattice (site, degree)       | $P_1$                                                                                                                             | Value                  |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 4.8.8 (3)                    | $c(4 - 3c)/16$                                                                                                                    | 0.08095858841632017693 |
| Union Jack, face centre (4)  | $9c^2(1 - c)/8$                                                                                                                   | 0.15402172338027135905 |
| Union Jack, corner (8)       | $\frac{1}{256}(9c - 2 - 4p)(6 - 7c - 4p)(7 - 9c - 4p)(81c^2 - 105c + 34 - 4p - 16p^2)^2$                                          | 0.02043343799324922073 |
| Star 3.12.12 (3)             | $7/100$                                                                                                                           | 0.07                   |
| Dice, rim (3)                | $3/16$                                                                                                                            | 0.1875                 |
| Dice, hub (6)                | $[(1 + 2r)(11 - 6r)/144]^2$                                                                                                       | 0.01261518916334840149 |
| Lieb, corner (4)             | $(\pi - 2)/(4\pi^3)$                                                                                                              | 0.00920452869398469827 |
| Lieb, edge (2)               | $1/4$                                                                                                                             | 0.25                   |
| Snub square (5)              | $\frac{8953}{3888} - \frac{281\sqrt{3}}{216} + (\frac{44\sqrt{3}}{27} - \frac{11}{4})p + (\frac{17}{36} - \frac{\sqrt{3}}{3})p^2$ | 0.06191455535342906165 |
| Checkerboard (6)             | $13(7 - 8p)(11 + 8p)(1 + 16p - 16p^2)/65536$                                                                                      | 0.05351525370368361765 |
| Triakis, lattice vertex (12) | $(6r - 1)(5 + 18r)^2(41 - 42r)^2/45562500$                                                                                        | 0.00359238258113540057 |
| Triakis, triangle centre (3) | $27/100$                                                                                                                          | 0.27                   |
| Kagome (4) [known [7]]       | $(5 + 3r)(17 - 6r)/1296$                                                                                                          | 0.07029827383269397637 |
| Laves graph (3)              | $1/12$                                                                                                                            | 0.083                  |
| Diamond (4)                  | $2/27$                                                                                                                            | 0.074                  |

PSLQ over  $\{\ln 2, \ln 3, \ln(1 + \sqrt{2}), \ln(2 + \sqrt{3}), \ln(\sqrt{2} + \sqrt{3}), G/\pi, \text{Cl}_2(\pi/3)/\pi, \text{Cl}_2(\pi/12)/\pi, \text{Cl}_2(5\pi/12)/\pi\}$  returned  $-3\mathbb{E} \ln \det L + 4\ln(2 + \sqrt{3}) + 40G/\pi = 0$  [25] with residual below  $10^{-58}$  at 60 digits. The numerical value 1.41086 is known [13]; we found no prior closed form.

The same genus-zero structure gives *algebraic* resistances (Table III):  $R_{\text{tri-sq}} = \frac{1}{2} - \sqrt{3}/18$  and  $R_{\text{tri-tri}} = 2\sqrt{3}/9$ . Foster's sum rule  $8(\frac{1}{2} - \sqrt{3}/18) + 2(2\sqrt{3}/9) = 4$  holds exactly. Planar duality,  $R(e) + R(e^*) = 1$  (an edge is in the spanning tree exactly when its dual edge is not in the dual tree), then gives the Cairo values  $\frac{1}{2} + \sqrt{3}/18$  and  $1 - 2\sqrt{3}/9$ . The remaining transfer currents at a snub-square vertex are  $Y(a, b) = 1 - \sqrt{3}/2$ ,  $Y(a, d) = \sqrt{3}/9$ ,  $Y(a', b') = (3 - \sqrt{3})/6$ ,  $Y(d, a') = \frac{1}{2} - 2\sqrt{3}/9$ ,  $Y(a, a') = 5\sqrt{3}/9 - 1 + 1/(2\pi)$  and  $Y(a, b') = \frac{1}{2} - \sqrt{3}/9 - 1/(2\pi)$ . Here  $d$  is the dimer arc;  $a, b$  each form a triangle with  $d$ ;  $a', b'$  form a triangle with each other;  $a'$  is collinear with  $a$  and  $b'$  with  $b$ .

Then

$$R_{\text{dimer}} = \frac{2\Phi}{\pi\sqrt{w(w+2)}}, \quad (16)$$

$$R_{\text{square}} = \frac{1}{2} - \frac{\sqrt{w}\Phi}{2\pi\sqrt{w+2}}, \quad (17)$$

$$z(w) = \frac{1}{\pi} \left[ 2G + \Phi \operatorname{arccosh}(w+1) + \text{Cl}_2\left(\frac{\pi}{2} + \Phi\right) + \text{Cl}_2\left(\frac{\pi}{2} - \Phi\right) \right]. \quad (18)$$

*Derivation.* Start from  $z(0) = 4G/\pi$ . Kirchhoff's relation  $\partial z/\partial w = \frac{1}{2}R_{\text{dimer}}(w)$  (the derivative of the log tree count with respect to a conductance is the sum of the corresponding edge resistances) then integrates via  $u + 1 = \sec \phi$  and

$$\int_0^\Phi \phi \sec \phi d\phi = \Phi \ln \tan\left(\frac{\pi}{4} + \frac{\Phi}{2}\right) - 2G + \text{Cl}_2\left(\frac{\pi}{2} + \Phi\right) + \text{Cl}_2\left(\frac{\pi}{2} - \Phi\right). \quad (19)$$

### B. Shastry–Sutherland network with arbitrary dimer conductance

Give the square bonds conductance 1 and the dimer bonds conductance  $w \geq 0$ , and let  $\Phi = \arccos(1/(w+1))$ .

*Special cases.* At  $w = 1$ ,  $\Phi = \pi/3$ , and  $\text{Cl}_2(\pi/6) + \text{Cl}_2(5\pi/6) = 4G/3$  recovers the snub-square values. At  $w = 2$ ,  $z(2) = 1.57336855075766435824$ , which coincides with the independently computed Union Jack entropy per vertex. As  $w \rightarrow \infty$ ,  $z = \frac{1}{2} \ln(2w) + 2G/\pi + o(1)$ .

### C. The king lattice

Take  $\mathbb{Z}^2$  with unit axis bonds and diagonal conductance  $w$ , and let  $\varphi = \arccos(2w)$ . Then

$$R_{\text{axis}} = \frac{\arccos(2w)}{\pi\sqrt{1-4w^2}}, \quad (20)$$

$$R_{\text{diag}} = \frac{1}{2w} - \frac{\arccos(2w)}{\pi w\sqrt{1-4w^2}}, \quad (21)$$

for  $w < \frac{1}{2}$ . For  $w > \frac{1}{2}$ , use the continuation  $\operatorname{arccosh}(2w)/\sqrt{4w^2-1}$ . The entropy is

$$z(w) = \frac{4G}{\pi} + \ln w + \frac{2}{\pi} \left[ \varphi \ln \tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right) + \operatorname{Cl}_2\left(\frac{\pi}{2} + \varphi\right) + \operatorname{Cl}_2\left(\frac{\pi}{2} - \varphi\right) \right], \quad (22)$$

and at  $w = 1$  (the plain king graph)

$$z_{\text{king}} = \frac{20G}{3\pi} = 1.94373936020545853426\dots, \quad (23)$$

with  $R_{\text{axis}} = \ln(2 + \sqrt{3})/(\pi\sqrt{3})$  and  $R_{\text{diag}} = \frac{1}{2} - R_{\text{axis}}$ . The determinant  $9 - (1 + x + x^{-1})(1 + y + y^{-1})$  belongs to a family of Mahler measures, so  $20G/(3\pi)$  may be implicit in that literature [15]. The elementary structure has a simple origin: after the exact  $k_2$  average and the substitution  $x = -1 + t^2$ , the remaining integrand has discriminant  $8w + (2w - 1)^2 = (2w + 1)^2$ , a perfect square.

*Corollaries.* (i) Eliminating each face centre of the Union Jack lattice with centre-bond conductance  $w$  (star-mesh) yields the king lattice with axis conductance  $1 + w/2$  and diagonal ratio  $w' = w/(4 + 2w)$ . At  $w = 1$  this reproduces  $R_{\text{corner-corner}} = c/2$ . (ii) The checkerboard lattice with crossed-diagonal conductance  $w$  has  $\det = 4(w + 1)[2(a + b) + (w - 1)ab]$ , with  $a = 1 - \cos(k_1 + k_2)$  and  $b = 1 - \cos(k_1 - k_2)$ . This is the king determinant with  $w' = (1 - w)/(2(1 + w))$ .

## V. RESISTOR NETWORK REDUCTIONS

### A. Adjacent resistances on Archimedean and Laves tilings

The 4.8.8 and Union Jack lattices are dual, and the entries in Table III satisfy  $R(e) + R(e^*) = 1$  identically. The 4.8.8 determinant factorizes as  $64 - 4(3 + \cos k_1)(3 + \cos k_2)$ , which is the origin of  $\arccos(1/3)$ . Owaidat computed Union Jack resistances numerically [16]; Cserti *et al.* gave exact mappings for kagome, dice and decorated lattices [17]. Kirchhoff solves on  $120 \times 120$ -cell tori, Richardson-extrapolated, agree with the table to about  $10^{-7}$ .

### B. Square lattice with arbitrary diagonal bonds

Give conductance  $w_1$  to bonds along  $e_1$ ,  $w_2$  along  $e_2$ ,  $w_+$  along  $e_1 + e_2$  and  $w_-$  along  $e_1 - e_2$ . Let  $\sigma = w_+ + w_-$

TABLE III. Exact adjacent-node resistances ( $1 \Omega$  resistors).  $c = \sqrt{2} \arccos(1/3)/\pi$ . Pairs related by planar duality satisfy  $R(e) + R(e^*) = 1$ .

| Tiling, edge              | $R$                                                 | Value               |
|---------------------------|-----------------------------------------------------|---------------------|
| Snub square, tri-sq       | $\frac{1}{2} - \frac{\sqrt{3}}{18}$                 | 0.40377495513506237 |
| Snub square, tri-tri      | $\frac{2\sqrt{3}}{9}$                               | 0.38490017945975051 |
| Cairo, dual of tri-sq     | $\frac{1}{2} + \frac{\sqrt{3}}{18}$                 | 0.59622504486493763 |
| Cairo, dual of tri-tri    | $1 - \frac{2\sqrt{3}}{9}$                           | 0.61509982054024949 |
| 4.8.8, square             | $\frac{1}{2} + \frac{c}{4}$                         | 0.63853160599489299 |
| 4.8.8, oct-oct            | $1 - \frac{c}{2}$                                   | 0.72293678801021400 |
| Union Jack, corner-corner | $\frac{c}{2}$                                       | 0.27706321198978599 |
| Union Jack, corner-centre | $\frac{2-c}{4}$                                     | 0.36146839400510700 |
| King, axis                | $\frac{\ln(2+\sqrt{3})}{\pi\sqrt{3}}$               | 0.24202564754285515 |
| King, diagonal            | $\frac{1}{2} - \frac{\ln(2+\sqrt{3})}{\pi\sqrt{3}}$ | 0.25797435245714485 |

and  $\Delta = 4w_+w_- - w_1^2$ . Then

$$R(e_1) = \begin{cases} \frac{2}{\pi\sqrt{\Delta}} \operatorname{arcsinh} \sqrt{\frac{\Delta}{(w_1 + \sigma)(w_1 + w_2)}}, & \Delta > 0, \\ \frac{2}{\pi\sqrt{-\Delta}} \operatorname{arcsin} \sqrt{\frac{-\Delta}{(w_1 + \sigma)(w_1 + w_2)}}, & \Delta < 0, \\ \frac{2}{\pi\sqrt{(w_1 + \sigma)(w_1 + w_2)}}, & \Delta = 0. \end{cases} \quad (24)$$

*Derivation.* At fixed  $x = \cos k_1$  the  $k_2$  average of  $1/\det$  is  $1/\sqrt{A^2 - M^2}$  with  $A^2 - M^2 = 4(1 - x)(p + qx)$ , where  $p = (w_1 + w_2 + \sigma)^2 - w_2^2 - (w_+ - w_-)^2$  and  $q = \Delta$ . The factor  $1 - x$  is the node at  $k = 0$ . Then  $R(e_1) = \pi^{-1} \int_{-1}^1 dx / \sqrt{(1 + x)(p + qx)}$ , and  $p - q = 2(w_1 + \sigma)(w_1 + w_2)$ . *Special cases.* The square lattice gives  $1/2$ ; the king lattice gives  $\arccos(2w)$  as above; and the triangular lattice ( $w_- = 0$ ) gives Kenyon's isoradial value [20],  $1/3$  in the isotropic case. Together with Foster's identity  $w_1 R(e_1) + w_2 R(e_2) + w_+ R(e_1 + e_2) + w_- R(e_1 - e_2) = 1$ , this also gives the diagonal resistances when  $w_+ = w_-$ . The formula agrees with direct 2D integration to  $\sim 10^{-26}$  for random conductances.

### C. Next-nearest-neighbour networks on bipartite lattices

Let the lattice be  $z$ -regular, bipartite and of girth at least 6, with unit nearest-neighbour bonds and conductance  $w$  on second neighbours. Since  $A^2 = zI + A_2$ ,

$$L = (z + z(z - 1)w)I - A - w(A^2 - zI), \quad (25)$$

$$\det = u(1 + 2zw + w^2u), \quad u = z^2 - |f(k)|^2. \quad (26)$$

The symmetry identity  $\mathbb{E}[\text{Ref}/\det] = \mathbb{E}[|f|^2/(z\det)]$  collapses the nearest-neighbour resistance to

$$R_{\text{NN}}(w) = 2\left(w + \frac{1}{z}\right) \mathbb{E}_k \left[ \frac{1}{1 + 2zw + w^2(z^2 - |f(k)|^2)} \right], \quad (27)$$

$$R_{\text{NNN}}(w) = \frac{2 - zR_{\text{NN}}}{z(z-1)w}. \quad (28)$$

*Honeycomb* ( $z = 3$ ,  $|f|^2 = 3 + 2S$ ,  $S = \cos k_1 + \cos k_2 + \cos(k_1 - k_2)$ ). Here  $\det = 2T(6w + 1 + 2w^2T)$  with  $T = 3 - S$ . Horiguchi's formula [21] with  $e = 3 + 1/w$  gives

$$m = \frac{16w^3(3w + 1)}{(2w + 1)^3(6w + 1)}, \quad (29)$$

$$R_{\text{NN}} = \frac{4(3w + 1)K(m)}{3\pi(2w + 1)^{3/2}\sqrt{6w + 1}}, \quad (30)$$

and  $R_{\text{NNN}} = (2 - 3R_{\text{NN}})/(6w)$ . At  $w = 1$ :  $m = 64/189$ ,  $R_{\text{NN}} = 16K(64/189)/(9\sqrt{21}\pi) = 0.21451969947398551853$  and  $R_{\text{NNN}} = 0.22607348359634057407$ . *Diamond* ( $z = 4$ ). Here

$$R_{\text{NN}} = \frac{4w + 1}{8w^2} P_{\text{FCC}}\left(3 + \frac{8w + 1}{4w^2}\right), \quad (31)$$

where  $P_{\text{FCC}}(t) = \mathbb{E}[1/(t - s)]$  is the FCC Green function in Joyce's closed form [22]. At  $w = 1$  ( $t = 21/4$ ),  $R_{\text{NN}} = 0.12347073437196593577$  and  $R_{\text{NNN}} = (1 - 2R_{\text{NN}})/(6w) = 0.12550975520934468$ . Three-dimensional Kirchhoff solves on  $12^3$ ,  $16^3$  and  $20^3$  tori give  $0.1234547 \rightarrow 0.1234640 \rightarrow 0.1234673$ .

## VI. DISCUSSION AND CONCLUSION

Three observations organize the results. First, the Majumdar–Dhar matrix *is* the transfer-current matrix, Eq. (2). Sandpile height probabilities, UST leaf and degree statistics, and resistances therefore all come from the same  $z \times z$  matrix at a vertex. Second, in three dimensions every needed resistance lies in the three-dimensional space spanned by  $\{1, W, 1/(\pi^2 W)\}$ . This is why  $P_1$  becomes a polynomial in the Watson integral, and why the determinants factor so strikingly. Third, in two dimensions, closed forms arise whenever the spectral curve has genus zero after removal of the node at  $k = 0$ . That gives algebraic or elementary resistances and Clausen-type entropies (snub square, Cairo, Shastry–Sutherland, king). Otherwise complete elliptic integrals appear (next-nearest-neighbour honeycomb).

Open questions include the higher height probabilities  $P_2, P_3, \dots$  in three dimensions, the ruby and snub-hexagonal lattices (for which PSLQ found no closed form in natural bases), and a conceptual explanation of the perfect-square factorization of the pyrochlore  $P_1$ .

### Appendix A: Added mass of a sphere touching a spherical container

A companion image-sum result is recorded here because it has the same flavour: an infinite image series collapses to a single special value. Consider a ball of radius  $a$  that touches, from inside, a fixed spherical tank of radius  $b > a$  filled with ideal (inviscid, incompressible, irrotational) fluid. The ball moves along the line of centres, and its added mass is  $M = C\rho\frac{4}{3}\pi a^3$ . With  $r = a/b$  and  $q = 1/(1 - r)$ ,

$$C(r) = \frac{3}{2} \sum_{k=0}^{\infty} \frac{1}{[1 + k(1 - r)]^3} - 1 = \frac{3}{2} q^3 \zeta(3, q) - 1, \quad (\text{A1})$$

where  $\zeta(s, q)$  is the Hurwitz zeta function. More generally, let the ball touch a sphere of *signed* radius  $b$  ( $b > 0$  a container,  $b = \infty$  a plane,  $b < 0$  an external ball of radius  $|b|$ ) and set  $\lambda = 1 - a/b$ . Then  $C = \frac{3}{2} \sum_{k \geq 0} (1 + k\lambda)^{-3} - 1$ . Special cases: a flat wall gives  $\frac{3}{2}\zeta(3) - 1 = 0.8030853\dots$  (known);  $r = \frac{1}{2}$  gives  $12(\zeta(3) - 1) - 1 = 1.4246828\dots$ ; and two equal touching balls give  $\frac{21}{16}\zeta(3) - 1 = 0.5776997\dots$

*Derivation.* The image of an axial dipole in a rigid sphere is an axial dipole at the inverse point with strength scaled by  $-(R/f)^3$  [23, 24]. Inverting about the contact point maps both spheres to parallel planes, so the image positions form an arithmetic progression. The strength ratios then telescope:  $A_k = A_0 b^3/[a + k(b - a)]^3$ . The added mass,  $-(\rho/U) \oint \phi n_x dS$  over the ball, is a sum of Hurwitz-zeta values.

*Verification.* We reconstructed the flow from the images and checked both boundary conditions at 2000 random surface points. Brute-force surface integration, with no zeta functions used, gives, for example, 1.424682623 ( $N = 6000$  images) against 1.424682838 for  $a = 1, b = 2$ . The truncation gap shrinks by a factor of about 4 per doubling of  $N$ . This result is probably a routine extension of classical image methods.

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[1] P. Bak, C. Tang, and K. Wiesenfeld, Self-organized criticality: An explanation of the  $1/f$  noise, *Phys. Rev. Lett.* **59**, 381 (1987).

[2] D. Dhar, Self-organized critical state of sandpile automaton models, *Phys. Rev. Lett.* **64**, 1613 (1990); Erratum, *Phys. Rev. Lett.* **64**, 2837 (1990).

- [3] S. N. Majumdar and D. Dhar, Height correlations in the Abelian sandpile model, *J. Phys. A: Math. Gen.* **24**, L357 (1991).
- [4] V. B. Priezzhev, Structure of two-dimensional sandpile. I. Height probabilities, *J. Stat. Phys.* **74**, 955 (1994).
- [5] N. Azimi-Tafreshi, H. Dashti-Naserabadi, S. Moghimi-Araghi, and P. Ruelle, The Abelian sandpile model on the honeycomb lattice, *J. Stat. Mech.* (2010) P02004, arXiv:0912.3331.
- [6] A. Poncelet and P. Ruelle, Sandpile probabilities on triangular and hexagonal lattices, *J. Phys. A: Math. Theor.* **51**, 015002 (2018), arXiv:1708.03493.
- [7] Q. Liu, J.-N. Herre, P. Baruah, S. Dreizler, C. Karrasch, W. Ruszel, and D. Schuricht, Universal correlations in the Abelian sandpile model, arXiv:2609.10352 (2026).
- [8] G. N. Watson, Three triple integrals, *Q. J. Math.* **os-10**, 266 (1939).
- [9] M. L. Glasser and I. J. Zucker, Extended Watson integrals for the cubic lattices, *Proc. Natl. Acad. Sci. USA* **74**, 1800 (1977).
- [10] G. Kirchhoff, Über die Auflösung der Gleichungen, auf welche man bei der Untersuchung der linearen Vertheilung galvanischer Ströme geführt wird, *Ann. Phys. Chem.* **72**, 497 (1847).
- [11] R. Burton and R. Pemantle, Local characteristics, entropy and limit theorems for spanning trees and domino tilings via transfer-impedances, *Ann. Probab.* **21**, 1329 (1993).
- [12] R. Lyons and Y. Peres, *Probability on Trees and Networks*, Cambridge Series in Statistical and Probabilistic Mathematics Vol. 42 (Cambridge University Press, Cambridge, 2016).
- [13] S.-C. Chang and W. Wang, Spanning trees on lattices and integration identities, *J. Phys. A: Math. Gen.* **39**, 10263 (2006), arXiv:cond-mat/0604589.
- [14] S.-C. Chang and R. Shrock, Some exact results for spanning trees on lattices, *J. Phys. A: Math. Gen.* **39**, 5653 (2006), arXiv:cond-mat/0602574.
- [15] A. J. Guttmann and M. D. Rogers, Spanning tree generating functions and Mahler measures, *J. Phys. A: Math. Theor.* **45**, 494001 (2012).
- [16] M. Owaidat, On the uniform tiling with electrical resistors, arXiv:1305.5839 (2013).
- [17] J. Cserti, G. Széchenyi, and G. Dávid, Uniform tiling with electrical resistors, *J. Phys. A: Math. Theor.* **44**, 215201 (2011), arXiv:1102.4537.
- [18] R. M. Foster, The average impedance of an electrical network, in *Reissner Anniversary Volume: Contributions to Applied Mechanics* (J. W. Edwards, Ann Arbor, 1949), pp. 333–340.
- [19] F. Y. Wu, Number of spanning trees on a lattice, *J. Phys. A: Math. Gen.* **10**, L113 (1977).
- [20] R. Kenyon, The Laplacian and Dirac operators on critical planar graphs, *Invent. Math.* **150**, 409 (2002).
- [21] T. Horiguchi, Lattice Green's functions for the triangular and honeycomb lattices, *J. Math. Phys.* **13**, 1411 (1972).
- [22] G. S. Joyce, On the cubic modular transformation and the cubic lattice Green functions, *J. Phys. A: Math. Gen.* **31**, 5105 (1998).
- [23] W. M. Hicks, On the motion of two spheres in a fluid, *Philos. Trans. R. Soc. Lond.* **171**, 455 (1880).
- [24] H. Lamb, *Hydrodynamics*, 6th ed. (Cambridge University Press, Cambridge, 1932).
- [25] H. R. P. Ferguson, D. H. Bailey, and S. Arno, Analysis of PSLQ, an integer relation finding algorithm, *Math. Comp.* **68**, 351 (1999).